PASSWORD SECURITY TOOL

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CUPRINS

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Application Description

The application is a desktop application developed in Java, using JavaFX for the graphical user interface and PostgreSQL for data storage.

The user can create an account by entering a username, date of birth, and a password. The password is analyzed in terms of security, and the application provides suggestions to improve it. Additionally, the user can generate

Technologies Used

The application is developed using the following technologies:

- **Java** – main programming language used for implementing the application logic
- **JavaFX** – used for building the graphical user interface
- **PostgreSQL** – database used to store user data
- **JDBC (Java Database Connectivity)** – used to connect the application to the database
- **Gradle** – build tool used for project management and dependency handling

Database

The application uses a PostgreSQL database to store user information.

The database contains a single table called **users**, which stores the following fields:

```
CREATE TABLE users (
   id SERIAL PRIMARY KEY,
   username VARCHAR(100) NOT NULL,
   birth_year VARCHAR(10),
   password_hash VARCHAR(255) NOT NULL
);

SELECT * FROM users;
```

The password is not stored in plain text, but as a hash value, ensuring a higher level of security for user data.

The connection to the database is made using JDBC.

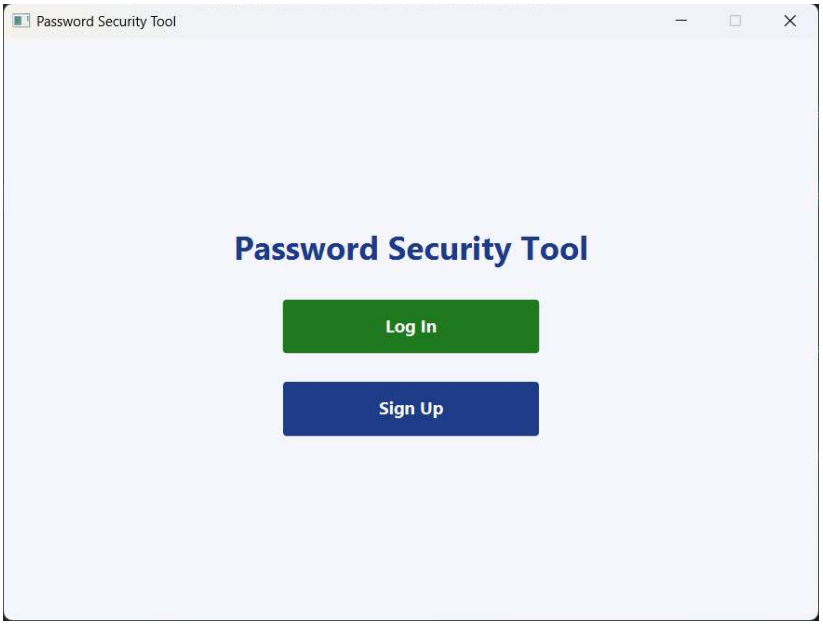
Functionalities

The application provides the following functionalities:

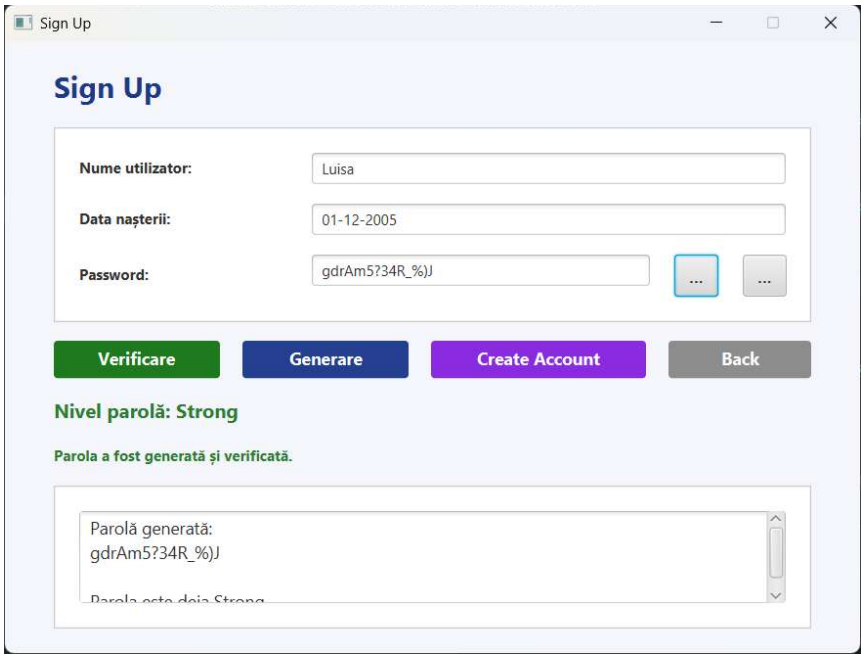
- **Sign Up** – allows the user to create a new account
- **Log In** – allows the user to log into the application
- **Password verification** – checks if the password is weak, medium, or strong
- **Password generation** – automatically generates a stronger password
- **Show password** – allows the user to see the entered password
- **Copy / Paste password** – allows easier use of the password in the application

User interface

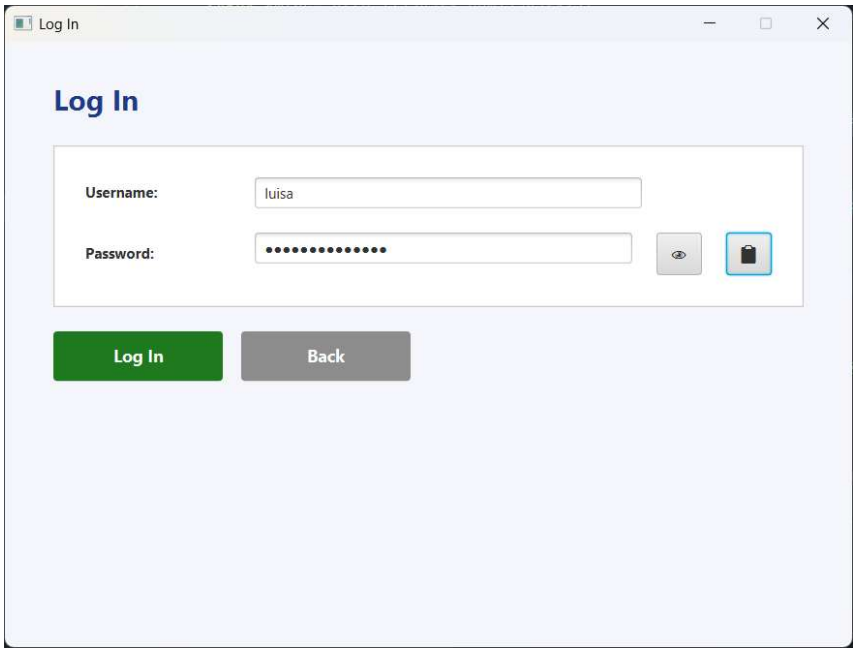
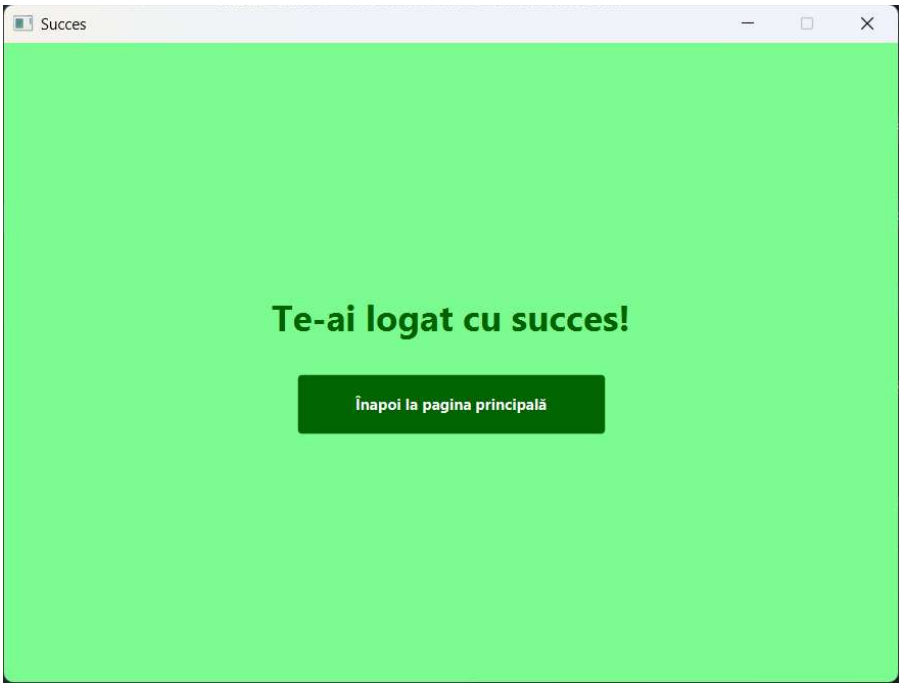
- Main Page
  - The main page allows the user to choose between logging in or creating a new account.



- Sign Up Page
  - The main page allows the user to choose between logging in or creating a new account.



- Log In page
  - The Log In page allows the user to authenticate using their username and password.



- Success Page
  - The Log In page allows the user to authenticate using their username and password.